

Relevant but Hard to Recover: An Empirical Study of AIGC Rewrites in Ranked Retrieval

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Abstract

AI-generated rewrites increasingly appear in retrieval corpora, where they may be derived from query-relevant passages while changing wording and source cues. Standard ranking evaluation measures whether a passage is useful for a query, but not whether multiple useful passages are independently sourced or recoverable as sharing the same source lineage. We present a controlled closed-pool empirical study over SciFact, FiQA, and FEVER, constructing same-lineage AIGC rewrite variants for query-positive passages and evaluating them with BM25, TF-IDF, BGE-small, and E5-small. AIGC rewrites often remain highly ranked and may outrank the original passage, while surface overlap drops and same-lineage recovery under text-only baselines becomes threshold-sensitive; for heavily paraphrased rewrites, surface-level same-lineage recovery is 0.023 under our text-only baselines, whereas embedding-based recovery changes from 0.935 to 0.378 across thresholds. Our claims concern relative ranking and same-lineage recoverability in controlled candidate sets rather than web-scale prevalence or production provenance systems. These results suggest that AIGC-aware IR evaluation should report source-lineage recoverability alongside ranking relevance.

CCS Concepts

• **Information systems** → **Evaluation of retrieval results**; *Retrieval models and ranking*; • **Computing methodologies** → *Natural language generation*.

Keywords

synthetic content, ranked retrieval, AIGC rewrites, source-lineage recoverability, text-only recovery baselines, provenance signals, evaluation gap

1 Introduction

AI-generated content (AIGC) rewrites are increasingly plausible retrieval candidates. They may appear as summaries, search-oriented rewrites, answer snippets, reposted content, or machine-edited variants of human-written passages. Such text can preserve topical usefulness for a query while changing surface form, wording, and source cues. This makes AIGC rewrites a natural object of study for information retrieval (IR) systems operating over mixed human-authored and synthetic corpora. This framing also aligns with SynthIR’s emphasis on AIGC-aware retrieval evaluation and source/provenance signals [19].

Traditional IR evaluation focuses on relevance and ranking utility. However, in AIGC-aware retrieval settings, a ranked list may contain several passages that are relevant but not independent because they descend from the same source lineage. A system can

therefore retrieve useful passages while obscuring whether the evidence comes from independent sources or from multiple rewrites of one source. This creates a source-lineage evaluation gap between what relevance-oriented metrics reward and what source-aware retrieval evaluation may need to measure.

We ask: when query-positive passages are rewritten by AIGC systems, do the rewritten variants remain attractive to retrieval models, and how much source-recovery signal remains available to simple text-only baselines? The question is not whether a rewrite is detectable as AI-generated text in isolation. Instead, we study whether a source-derived passage can retain retrieval value while becoming less recoverable as a same-source variant under token overlap, character shingles, embedding similarity, and simple recovery baselines.

We construct controlled same-lineage rewrite variants over real query-positive passages from SciFact, FiQA, and FEVER. For each original passage E_0 , we generate four AIGC variants ranging from light rewriting to search-oriented rewriting, summarization, and heavy paraphrasing. We evaluate BM25, TF-IDF, BGE-small, and E5-small in candidate pools that contain E_0 – E_4 and real distractor passages from the same dataset. We jointly measure retrieval exposure, source-recovery proxies, and surface/embedding recovery baselines.

Contributions. We make four contributions: (1) we formulate AIGC rewrite-induced source-lineage recoverability loss under simple text-only baselines as a measurement problem for ranked retrieval; (2) we construct controlled same-lineage rewrite variants over real IR datasets; (3) we show that source-derived rewrites can remain highly ranked while surface recovery signals weaken; and (4) we show that surface deduplication is limited and embedding-based recovery is threshold-sensitive. Our claims concern relative ranking and same-lineage recovery in controlled closed-pool candidate sets, rather than web-scale prevalence or production provenance systems.

2 Related Work

IR evaluation and retrieval benchmarks. Classical IR evaluation emphasizes ranked retrieval effectiveness, with metrics such as nDCG and lexical baselines such as BM25 remaining central reference points [6, 17]. Modern benchmarks such as MS MARCO and BEIR evaluate retrievers across heterogeneous tasks and domains [14, 20]. We use established IR datasets and retrievers as measurement infrastructure, but our object of study is different: we measure how same-lineage AIGC rewrites affect ranking exposure and source-recovery proxies in controlled per-query pools built from SciFact, FiQA, and FEVER [12, 21, 22].

Synthetic text in retrieval. Prior work has used generated text to improve retrieval, including document expansion, synthetic

queries, and generative pseudo-labeling [1, 15, 23]. These methods show that synthetic text can be useful for ranking. Recent work on LLM-generated text and term-based retrieval argues that BM25-style systems respond to query-document term-distribution alignment rather than showing an inherent source preference [5]. Retrieval Collapse studies corpus-level AI-content pollution and source-diversity erosion in web-style retrieval pipelines [26]. In contrast, we do not study retrieval improvement or web-scale contamination; we construct same-lineage AIGC rewrites and ask whether text-only source-recovery signals still connect ranked variants to their original passage.

Paraphrasing, embeddings, and source recovery. Krishna et al. show that paraphrasing can evade AI-text detectors while retrieval over stored generations can recover semantically similar sources [10]. Dense retrieval and sentence embeddings provide semantic matching signals beyond lexical overlap [8, 16], and we use BGE-small and E5-small as representative dense retrievers [11, 13, 24]. We also draw on near-duplicate signals such as token overlap, character shingles, and thresholded embedding similarity [2, 3]. Our focus differs from detector robustness and embedding retrieval alone: we compare what ranking metrics reward with what simple text-only recovery baselines can still link to the original source lineage.

Provenance and synthetic-content signals. Watermarking, detection, and content-provenance systems address whether generated content can be identified or accompanied by metadata [4, 9, 18]. Interactive provenance systems such as Traceable Text link generated summary spans back to source passages at phrase level [7]. Cross-modal work on invisible relevance bias shows that generated images can change text-image retrieval preferences even under matched relevance [25]. Our study is complementary: we do not assume provenance metadata is available, and we measure the weaker but retrieval-relevant question of whether text-only source-recovery signals remain available for AIGC rewrites in ranked retrieval results.

3 Problem Setup

For each query-positive pair (q, E_0) , q is a retrieval query and E_0 is an original human-written passage judged relevant to q . Starting from E_0 , we generate four AIGC rewrite variants: E_1 light rewrite, E_2 search-friendly rewrite, E_3 abstract summary, and E_4 heavily paraphrased same-lineage rewrite. The five passages E_0 – E_4 are distinct retrievable texts, but they are not independent sources: all variants share the same `source_lineage_id`. The retriever observes only passage text, not lineage metadata.

The experimental question is whether ranking exposure and source recovery diverge. The ranking view asks whether E_1 – E_4 appear in top results and whether they outrank E_0 . The source-recovery view asks whether those same variants can still be linked to E_0 using token overlap, character shingles, thresholded embedding similarity, or simple recovery baselines. In this paper, “harder to recover” means less recoverable under these text-only proxies, not forensic untraceability.

For each query, the candidate pool contains E_0 – E_4 plus 50 real distractor passages from the same dataset. This controlled abstraction models a common form of AIGC rewrite reuse: several query-useful

passages may descend from the same source while appearing as separate candidates to the retriever. Our claims are therefore about relative ranking and same-lineage recovery in this controlled setting, not about all web corpora or production provenance systems.

4 Experimental Design

Datasets and rewrite variants. We use SciFact, FiQA, and FEVER. For each dataset, we sample 200 query-positive passage pairs from existing qrels or positive passages; each query pool contains E_0 – E_4 plus 50 real distractors from the same dataset, and E_0 is the real human-written positive passage rather than a template. For each E_0 , we generate E_1 – E_4 and assign all variants the same source-lineage identifier. Generation uses Qwen2.5-1.5B-Instruct with greedy decoding, a 310-token input truncation, and edit-specific output caps; the fixed templates instruct the model to preserve factual claims, named entities, and numbers across four rewrite intents: E_1 light rewriting, E_2 search-friendly rewriting, E_3 abstract-style summarization, and E_4 heavy paraphrasing. Table 1 summarizes the controlled closed-pool setup.

Table 1: Controlled closed-pool setup. Each query pool contains the original query-positive passage (E_0), four same-lineage AIGC rewrites (E_1 – E_4), and 50 real distractors from the same dataset.

Dataset	Query-positive pairs	Same-lineage docs	Distractors/query
SciFact	200	E_0 – E_4	50
FiQA	200	E_0 – E_4	50
FEVER	200	E_0 – E_4	50

Candidate pools and retrievers. The closed-pool construction tests whether rewrites rank among real distractors rather than only among synthetic alternatives. We evaluate BM25 and TF-IDF as lexical retrievers, and BGE-small and E5-small as dense retrievers. All reported results are computed from saved retrieval runs and traceability metrics; no new method is introduced. The setup is intentionally small and closed: it is a stress test for whether ranking and source-lineage proxies diverge, not an estimate of web-scale prevalence.

Automatic fidelity checks. Because rewriting may change specificity or answerability, we do not treat source-derived rewrites as human-validated relevance judgments. We run automatic checks for entity preservation, number preservation, and query-anchor retention, and use a local judge when available. We also performed a small manual spot-check over sampled rewrites from the main three-dataset setting. The check was used only as a qualitative sanity check and further suggested that factuality preservation remains a limitation; therefore, we interpret the main results as source-derived retrieval exposure rather than human-validated relevance or semantic equivalence.

5 Metrics

We aggregate over query-condition pairs, comparing each rewrite condition E_i with the original passage E_0 inside the same candidate pool.

Retrieval relevance. **EiTop10** measures whether E_i appears in the top 10. **EiBeatsOriginal** measures whether E_i ranks above

E_0 . **RankLift** is the rank improvement of E_i relative to E_0 . These are ranking/exposure metrics, not independent human relevance judgments for the rewrites; the rewrites are source-derived from query-positive originals.

Text-only source-recovery proxies. **Token Jaccard** and **Char-5 Jaccard** measure surface similarity between E_i and E_0 . **Embedding cosine** measures semantic similarity. We also report **LowSurf-HighSem@10**, which measures top-10 rewrites with low character-level overlap but high embedding similarity to the original passage. It is defined as the mean of $1[\text{rank}(E_i) \leq 10 \text{ and } \text{Char5J}(E_i, E_0) \leq 0.30 \text{ and } \text{EmbCos}(E_i, E_0) \geq 0.75]$. The thresholds are descriptive operating points chosen to mark low character-level overlap and retained embedding-level similarity; they are not tuned attribution decision boundaries, and Table 3 reports sensitivity across stricter embedding thresholds.

Source recovery. We evaluate exact matching, token Jaccard, character 5-gram Jaccard, and embedding cosine thresholds. **SameLineageDetectionRate** is recall on positive pairs (E_i, E_0) : a baseline links a rewrite to its original if the corresponding exact-match or similarity threshold is passed. **FalseSplitRate** is $1 - \text{recall}$ on these same-lineage pairs, and **FalseMergeRate** is the false-positive rate on sampled different-lineage negatives.

6 Results

6.1 Finding 1: AIGC Rewrites Remain Competitive in Ranked Retrieval

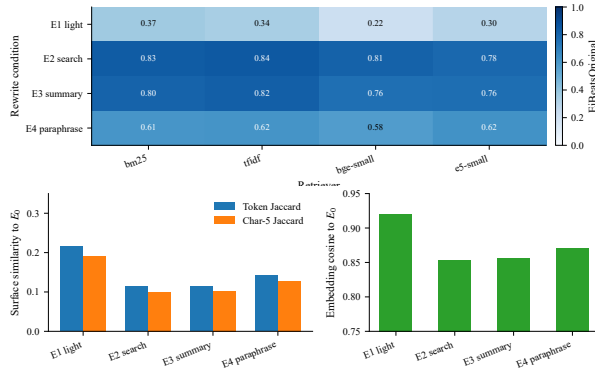


Figure 1: Main empirical pattern. Top: rewrite variants often outrank the original passage across retrievers. Bottom left: surface similarity to E_0 drops under rewriting. Bottom right: embedding similarity to E_0 remains high.

Across datasets and retrievers, AIGC rewrites frequently remain in the top 10. The mean EiTop10 values are 0.991 for E_2 , 0.991 for E_3 , and 0.976 for E_4 . Rewrites also often outrank the original passage: for example, EiBeatsOriginal is 0.840/0.865 for E_2/E_3 on FEVER with BM25, 0.875/0.750 on FiQA with BGE-small, and 0.880/0.875 on SciFact with BGE-small. Thus, rewritten variants can be attractive to both lexical and dense retrievers under our controlled candidate pools. E_2 uses a search-friendly prompt, so part of its ranking advantage is expected; we use it as a stress-test condition rather than as an unbiased naturally occurring rewrite. We do not claim the effect is uniform. Figure 1 summarizes the main empirical pattern:

rewrites remain competitive in ranking, surface similarity drops, and embedding similarity remains high. Bootstrap intervals in the expanded run show that some large examples are stable; for FEVER/BM25, EiBeatsOriginal is 0.840 [0.785, 0.885] for E_2 , 0.870 [0.825, 0.915] for E_3 , and 0.620 [0.550, 0.688] for E_4 . The implication is that relevance-oriented ranking can treat source-derived rewrites as useful candidates, even when those candidates are not independent sources.

6.2 Finding 2: Ranking Exposure and Surface Recovery Diverge

Table 2 shows that AIGC rewrites often satisfy a low-surface/high-semantic retrieval pattern.

Table 2: Main results averaged across SciFact, FiQA, FEVER and four primary retrievers. Query-positive-derived rewrite variants often receive ranking exposure while surface similarity to E_0 drops. LS-HS@10 abbreviates LowSurf-HighSem@10.

Cond	Top10	Beats	Lift	LS-HS@10	TokJ	Char5J	EmbCos
E_1 light	0.944	0.302	-1.137	0.847	0.218	0.192	0.919
E_2 search	0.991	0.812	1.755	0.871	0.114	0.101	0.853
E_3 summary	0.991	0.777	1.604	0.878	0.113	0.102	0.855
E_4 para.	0.976	0.596	0.378	0.880	0.143	0.128	0.870

For E_2 and E_3 , token Jaccard is near 0.11 and Char-5 Jaccard is near 0.10, while embedding cosine remains above 0.85. The pattern is not a monotone edit-depth curve: E_4 is not the lowest-overlap condition under every proxy. This is expected because the labels are rewrite intents rather than an ordinal severity scale; summaries and search-style rewrites can remove more surface text than a paraphrase. However, E_4 remains semantically close under embedding similarity and is frequently retrieved, while being challenging for surface-level source-recovery baselines. The phenomenon should not be read as a monotonic edit-depth effect. Instead, the results suggest that ranking exposure and text-only source-recovery proxies can move differently under AIGC rewriting. The implication is that surface source cues can degrade even when semantic similarity remains high enough for retrieval models to keep the rewrite competitive.

6.3 Finding 3: Surface Recovery Is Limited; Embedding Recovery Is Threshold-Sensitive

For the heavily paraphrased E_4 condition, same-lineage recovery under text-only baselines is limited. The combined surface-level detection rate is 0.023; exact matching detects none of the E_4 rewrites, token Jaccard at threshold 0.5 detects 0.013, and character 5-gram Jaccard at threshold 0.3 detects 0.032. Embedding-based recovery is stronger but threshold-sensitive: detection changes from 0.935 at threshold 0.75 to 0.683 at 0.85 and 0.378 at 0.90. The expanded run adds BM25-near-duplicate, MinHash, and SimHash-style surface baselines; their E_4 detection remains near zero at the tested operating points, while embedding thresholds remain the only high-recall text-only recovery signal. The implication is that source recovery requires explicit design choices rather than default surface deduplication. Table 3 summarizes text-only same-lineage recovery for the E_4 condition.

Table 3: Text-only source recovery for E_4 . Surface matching recovers few same-lineage rewrites, while embedding recovery is sensitive to threshold choice.

Recovery baseline for E_4	Detect.	False split
Exact match	0.000	1.000
Token Jaccard ≥ 0.5	0.013	0.987
Char-5 Jaccard ≥ 0.3	0.032	0.968
Embedding cosine ≥ 0.75	0.935	0.065
Embedding cosine ≥ 0.85	0.683	0.317
Embedding cosine ≥ 0.90	0.378	0.622

6.4 Summary

The controlled study shows a consistent pattern: AIGC rewrites derived from query-positive passages can rank highly while becoming harder to recover with the studied text-only proxies. This exposes a possible evaluation gap between relevance-oriented IR evaluation and source-lineage recoverability under simple text-only baselines in a controlled measurement setting. We additionally ran an expanded robustness check; we use it only as sensitivity evidence and keep the main claim focused on the three-dataset controlled setting. This separation between ranking exposure and source recovery is the central measurement signal of the paper.

7 Discussion and Limitations

Implications for AIGC-aware IR evaluation. Our results suggest that AIGC-aware IR evaluation should report both relevance utility and source-lineage recoverability. A relevant passage may be useful for answering a query, but it may not provide independent evidence if it is a rewrite of another retrieved source. This distinction matters because relevance metrics and source-lineage metrics answer complementary questions.

Scope and limitations. Our claims concern relative ranking and same-lineage recovery in controlled closed-pool candidate sets, not web-scale prevalence, forensic untraceability, or the failure of production provenance systems. The study uses three datasets and one local rewrite model. Automatic fidelity checks and a small manual spot-check suggest that factuality preservation remains a limitation, so we interpret the results as source-derived retrieval exposure rather than human-validated relevance or semantic equivalence. Production systems may use stronger provenance metadata, cross-encoder alignment, editorial records, or platform-level source signals beyond the baselines studied here.

Benchmark design. Future SynthIR-style benchmarks could pair relevance judgments with explicit source-lineage annotations, making source lineage a measurable axis of AIGC-aware IR evaluation. Such annotations would allow evaluators to distinguish original human-written sources, AIGC rewrites derived from those sources, and unrelated passages that independently support the same query. This is the distinction that standard relevance labels often leave implicit: a ranked list may contain several relevant passages while providing less independent evidence than it appears to provide.

Reporting recommendation. For workshop-scale measurement studies, we recommend reporting source-lineage recoverability as an auxiliary diagnostic alongside standard ranking metrics. The goal is not to replace relevance evaluation or to treat all rewritten content as low quality. Instead, it is to make the evaluation target

explicit: a system can be good at ranking useful evidence while still making it difficult to tell whether multiple ranked items share a source lineage. Table 3 illustrates why this should be reported at several operating points rather than at a single threshold. Lower embedding thresholds recover more same-lineage rewrites but may increase false merges in broader settings, whereas stricter thresholds reduce merges but split more same-lineage variants. Surface overlap baselines are easy to interpret, but they miss heavily paraphrased rewrites under the tested settings.

What remains open. The next step is to combine this controlled ranked-retrieval setup with richer source metadata and human validation. Human judgments could separate factual preservation, answer usefulness, and source independence, which are deliberately conflated only minimally in this measurement study. Production systems may also expose editorial records, cryptographic credentials, or platform-level provenance signals that are unavailable to text-only baselines. These stronger signals should be evaluated directly rather than inferred from surface overlap. The present study therefore identifies a measurement gap for SynthIR-style tasks: relevance can remain high even when source-lineage recovery under simple text-only baselines becomes fragile.

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